

China green tech and its industrial policy

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Abstract: This paper examines China's emergence as the dominant global manufacturer of green technology, a development originally driven by its pressing domestic need to address environmental degradation and energy security. By implementing a comprehensive industrial strategy anchored in the strategic acquisition of foreign technology, extensive state subsidization, and the cultivation of a hyper-competitive domestic market, China has achieved unparalleled production capacity. Consequently, this capacity now significantly outstrips global demand, transforming China into the primary green technology supplier for the world. While this dynamic exports deflationary pressures that make global decarbonization more affordable, it simultaneously presents a formidable competitive challenge to foreign industries. The paper concludes by exploring strategic pathways for developing nations. It argues that the Global South must form robust regional alliances to aggregate market demand, leverage critical mineral resources, and build complementary manufacturing capabilities to successfully navigate this new, China-centric industrial landscape.

Keywords: green technology, industrial policy, decarbonization, energy transition, state subsidies

JEL classification: O25, Q42, F21

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1 Introduction

China's emergence as a global industrial hegemon in green technology is not an accidental phenomenon, nor can it be understood merely as an export-oriented strategy. Rather, its current dominance is the direct and calculated consequence of a pressing domestic imperative to address its own monumental environmental and energy security challenges. As the world's largest energy consumer and carbon emitter, China's long term economic development pathway has become inextricably linked to its capacity for technological innovation and mass deployment in the green sector. The strategic necessity of transition, driven by the need to mitigate severe negative externalities and secure energy independence, has in turn fostered a vast and state supported domestic market. This protected ecosystem has functioned as a crucible for forging the industrial capacity that now commands the global stage.

2 Related literature

Given China's importance on global climate challenges but also the perceived success about decarbonization policies, a wealth of literature exists with important policy recommendations for the rest of the world. As regards China's decarbonization efforts, the literature confirms their foundation in high-level, long-term national strategy. The 'Double Carbon' Policy, which aims for a carbon emissions peak before 2030 and carbon neutrality by 2060, is identified as a critical driver that has significantly raised the strategic priority of green innovation (Shi 2023). This builds on a history of integrating environmental goals into the nation's Five-Year Plans, with the 11th Five-Year Plan being an early, notable green economy and industry development plan (Korabayev and Korabayev 2023).

Key studies analyse China's complex and multi-layered policy toolkit, which combines broad national directives with targeted measures like financial incentives and localized pilot programs. The New Energy Demonstration City (NEDC) initiative, for instance, creates local platforms for green innovation. Empirical research indicates the NEDC policy boosts corporate green innovation and reduces firm-level energy intensity via technological upgrades (Cheng, Yu, and Zhang 2023). However, the policy's impact is not uniform, showing more pronounced effects for resource-based cities and cities under greater fiscal pressure. Critically, while the program fosters innovation within pilot zones, it can also create negative spillover effects by concentrating resources in those areas (Ma 2025).

Financial levers represent another core component of state strategy. Green Credit Policies (GCPs), which use market-based incentives to direct capital towards green projects, have received a mixed verdict from researchers. One study found that GCPs tended to restrict innovation in heavily polluting enterprises due to heightened financial constraints (Sun et al. 2025). Conversely, other research suggests GCPs successfully facilitated green innovation, leading to a reported 15% increase for heavily polluting firms compared to control groups. This positive effect is sometimes mediated by Corporate Social Responsibility (CSR), as the policy can encourage CSR engagement that subsequently promotes green innovation (He et al. 2025).

Direct subsidies are widely used to underwrite R&D risks; however, their efficacy is intensely debated. Research on the New Energy Vehicle (NEV) industry, which has been a significant beneficiary of governmental support, demonstrates that subsidies have historically fallen short in enhancing. In

certain instances, even hampered industry's fundamental innovation capacity. This deficiency is attributed to a prioritization of production over innovation and the prevalence of rent seeking behaviour (Yang 2023). This finding aligns with broader analyses suggesting that single instruments often have limited effectiveness, highlighting the need for a coordinated policy mix (Veugelers 2014). Beyond domestic policy, the international activities of Chinese firms provide another channel for innovation. Outward Foreign Direct Investment, particularly under the Belt and Road Initiative, has been shown to promote firms' green technological innovation by easing financial constraints and stimulating R&D (Chen, Chen, and Xu 2025).

Despite being a critical component of the worldwide energy transition, the global spread of green technologies faces substantial barriers, notably in Asia. A significant challenge is the intellectual property rights framework, which while designed to foster innovation, can impede technology transfer to developing nations (Kariyawasam and Tsai 2018). Furthermore, many developing countries are trapped in a cycle where high upfront capital costs and inadequate policy support hinder their shift from fossil fuels to clean energy (Peimani 2018). These financial and regulatory hurdles are compounded by a digital divide across the Indo-Pacific, where insufficient capital, a technology skills gap, and poor local infrastructure create significant barriers to green technology adoption and diffusion (Maji and Laha 2021).

In Southeast Asia, despite abundant potential for sustainable energy, slow technology adoption is hampered by substantial financing requirements, complex energy governance, and financial risks (Nepal, Phoumin, and Khatri 2021). Similarly, South Asian nations grapple with the dual challenge of meeting escalating energy demands while curbing greenhouse gas emissions, a task hindered by heavy reliance on fossil fuels and various structural barriers to green energy development (Quadrat-Ullah 2023).

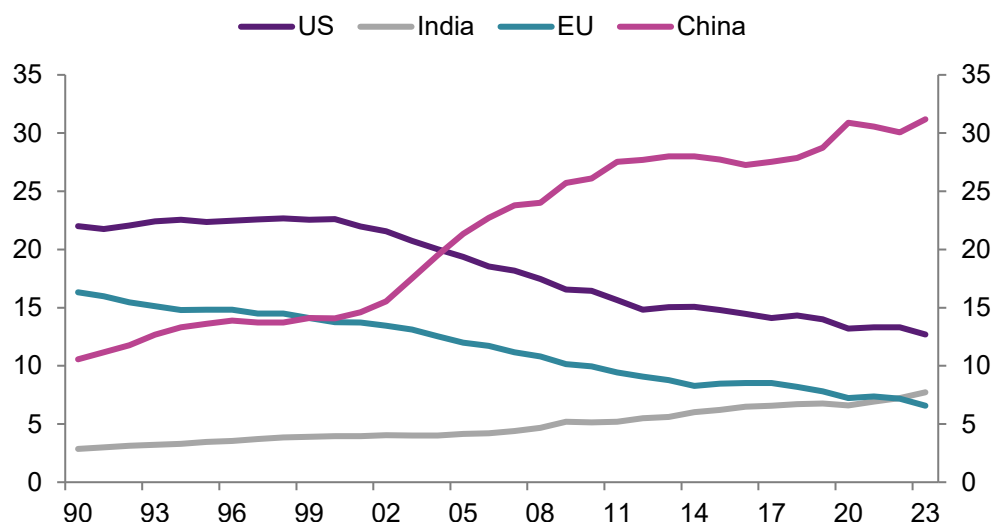
In this paper, we aim at contributing to the existing literature on China's green industrial policy to draw lessons for emerging countries which are facing similar decarbonization challenges. Section 3 focuses on China's distinct dual role as both a primary consumer and producer of green technology , while Section 4 calculates the required global installations against China's production capacity to quantify its manufacturing dominance. Section 5 deconstructs the multifaceted industrial strategy that enables this capacity, analysing the three core pillars of strategic technology acquisition, extensive state subsidies, and state-managed hyper-competition. Section 6 evaluates the broad economic consequences of China's dominance, including its deflationary effects on the cost of the global energy transition and the formidable competitive challenge it poses. Finally, Section 7 explores strategic pathways for the Global South, arguing that developing nations must form regional alliances to aggregate market demand, leverage critical mineral resources, and build complementary manufacturing capabilities to navigate this new industrial landscape.

3 Zooming into China's key dual role in decarbonization

The foundational driver for this strategic pivot is the sheer scale of China's carbon footprint and the associated economic liabilities. Accounting for over 30% of global CO₂ emissions, the nation faces immense internal pressures from the tangible costs of environmental degradation and increasing external pressure from the international community. From a macroeconomic perspective, high emissions trajectory represents a significant long-term risk, exposing the economy to the future

imposition of global carbon pricing regimes, such as carbon border adjustment mechanisms (CBAMs), and the severe physical impacts of climate change (Wang and Hao 2012). Consequently, the green transition has been framed within Beijing's economic planning as a national strategic priority, essential for mitigating these profound economic risks and ensuring long term energy and national security.

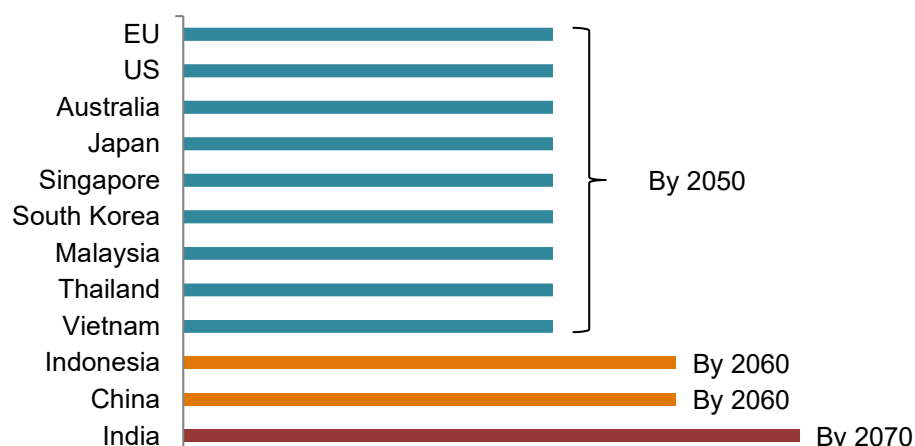
Figure 1: Global emission share (%)



Source: Natixis, Energy Institute

While China has been steadily building its green tech capabilities for years, the latest and most powerful catalyst for growth arrived with the announcement of the 'Dual Carbon' goals in 2020. By pledging to peak carbon emissions before 2030 and achieve carbon neutrality by 2060, the Chinese government created a clear and powerful long term demand signal (Shi 2023). This ambitious agenda unleashed a new wave of growth momentum across the sector.

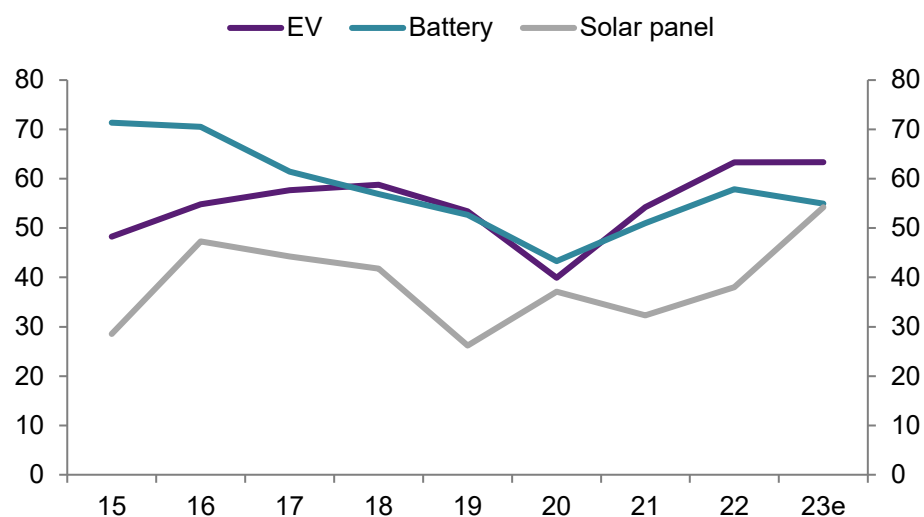
Figure 2: NDC on net zero emission time



Source: Natixis, UNFCCC NDC Registry.

The 'Dual Carbon' objectives have transformed China into the world's most significant single market for green technologies. The sheer scale of domestic demand required to meet these targets is enormous. As a result, China now accounts for the majority of global demand in key green sectors, including 64% for electric vehicles (EVs) and over half for batteries and solar panels.

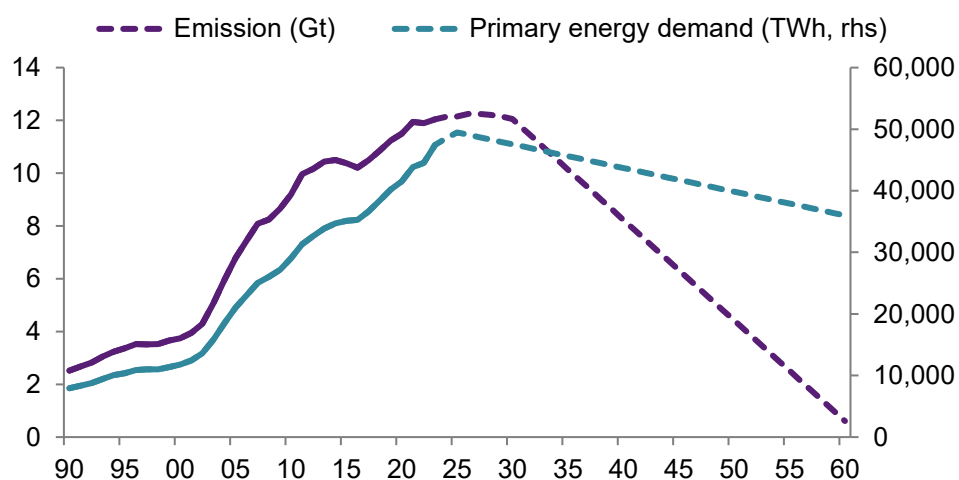
Figure 3: Global share of China's demand (%)



Source: Natixis, China Association of Automobile Manufacturers, SNE Research, China Photovoltaic Industry Association, CEIC.

Looking ahead, the path to carbon neutrality requires a fundamental restructuring of China's entire energy system. Projections indicate that no fossil fuels must constitute at least 80% of the total energy mix by that year. To achieve this, China's overall energy demand must be managed to peak in the near future, while the installed capacity of renewable technologies must continue to expand at a historic pace. In the long term, policy driven demand pipeline ensures that the domestic market will remain the primary engine of China's green tech sector for decades to come.

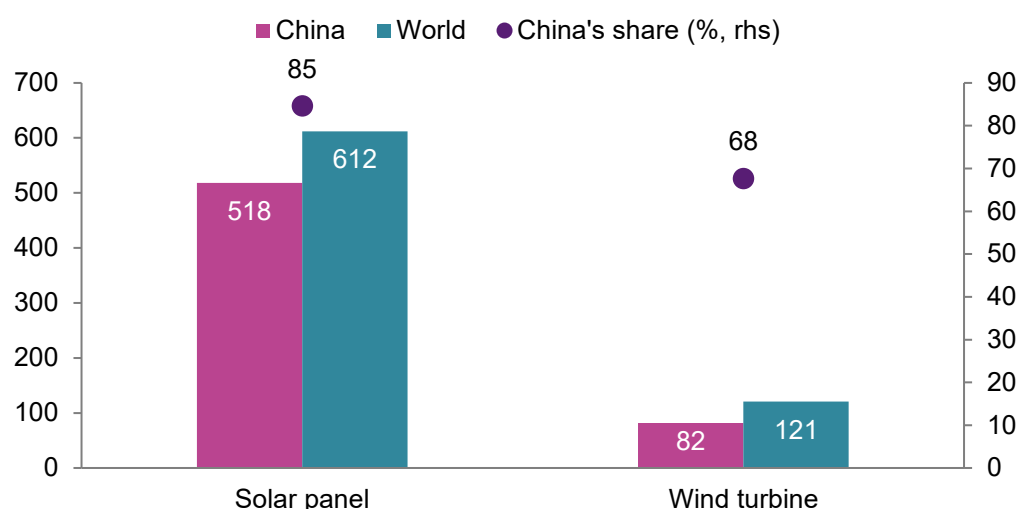
Figure 4: China's projected emission and energy demand towards dual-carbon



Source: Natixis, National Bureau of Statistics, Energy Institute.

China's manufacturing dominance is a direct outcome of cultivating its massive domestic market. The immense volume of internal demand allows firms to achieve unparalleled economies of scale and process optimization through aggressive learning by doing. This state underwritten ecosystem systematically drives down production costs, establishing a formidable global competitive advantage. China manufactures 85% of the world's solar panels and 68% of its wind turbines, figures that signify a near monopolistic control over the solar supply chain and a commanding position in wind energy.

Figure 5: Renewable equipment production, 2023 (GW)



Source: Natixis, China Photovoltaic Industry Association, Global Wind Energy Council.

Ultimately, the scale of China's domestic green transition is now a globally significant macroeconomic factor. The vast installation of renewable capacity, initially conceived to address its internal energy and environmental imperatives, has created an industrial base with worldwide influence. The process of solving its own problems has equipped China with the capacity to fundamentally shape the economics of the global energy transition.

4 China's increasing role in manufacturing for the rest of the world

4.1 Demand from the rest of the world

Having established an unparalleled industrial base to serve its domestic needs, China's green technology sector is now pivotal to the global energy transition. Its role is rapidly evolving from a self-sufficient ecosystem into the primary manufacturer for the rest of the world. This section analyzes the scale of China's manufacturing capacity in relation to global demand. We investigate whether the world's decarbonization efforts are becoming structurally dependent on Chinese production.

4.2 Pinning down the capacities needed

To quantify the global need for green technology, we first establish a baseline for demand. First, we compile various forecast scenarios of China's and the world's energy baskets up to 2050 (Raimi et al. 2024). Based on their demand for renewables, we then calculate the number of installations needed to fulfil such demand. From the 9 scenarios we collected, 4 were shortlisted by 3 criteria: 1) scenario complies with their announced climate pledges, 2) China reaches 25% renewable energy by 2030, and 3) China reaches 80% renewables by 2060 (we use linear extrapolation for scenarios covering only up to 2050). Table 1 summarizes the details below. This methodology provides a robust and credible framework for projecting the required pace of renewable energy deployment for both China and the rest of the world (ROW).

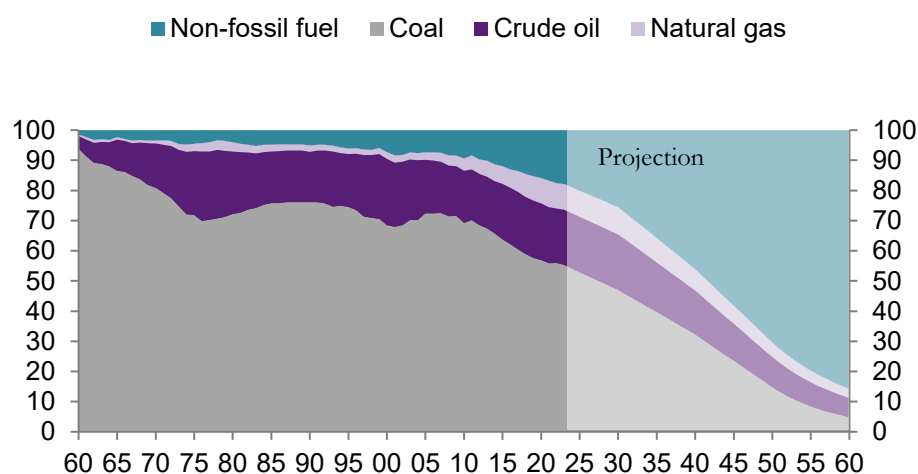
Table 1: Energy demand scenarios

Forecast Body	Scenario	Non-Fossil Fuel Share in Energy Mix for China (%)		Compliant with China's goal ?	Non-Fossil Fuel Share in Energy Mix for ROW (%)	
		2030	2050		2030	2050
International Energy Agency	Announced Pledges Scenario	26	70	Yes	32	61
	Stated Policies Scenario	22	46	No	28	39
British Petroleum	Net Zero	31	79	Yes	34	73
	Accelerated	25	71	Yes	30	59
	New Momentum	16	42	No	27	35
Enerdata	EnerGreen	31	79	Yes	43	77
	EnerBlue	19	60	No	34	58
	EnerBase	17	24	No	26	34

Source: Natixis, International Energy Agency, British Petroleum, Enerdata

We pin down the projected energy baskets for China and the world by taking the scenario with highest renewable energy demand forecast. The projected energy transition paths for China and the ROW diverge significantly. China's energy mix, historically dominated by coal, is projected to undergo a rapid and fundamental transformation. Coal's share will decline steeply and consistently, while non-fossil fuels, primarily solar and wind, are set to become the dominant energy source by mid-century.

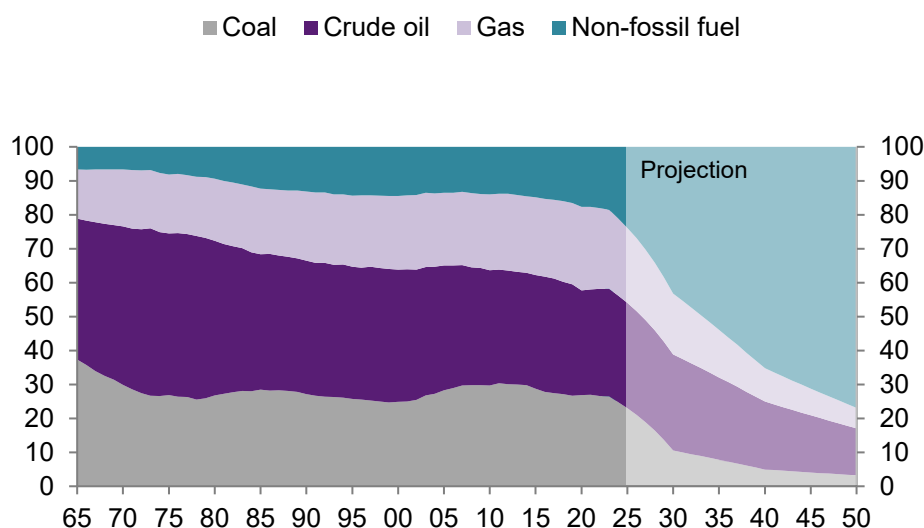
Figure 6: China primary energy mix (%)



Source: Natixis, National Bureau of Statistics, International Energy Agency.

In stark contrast, the ROW's energy profile shows a much more gradual transition. Its reliance on a diverse mix of crude oil, natural gas, and coal creates significant structural inertia. This makes the global decarbonization pathway far more challenging, requiring a much more aggressive and immediate shift in energy consumption than in China to meet climate targets.

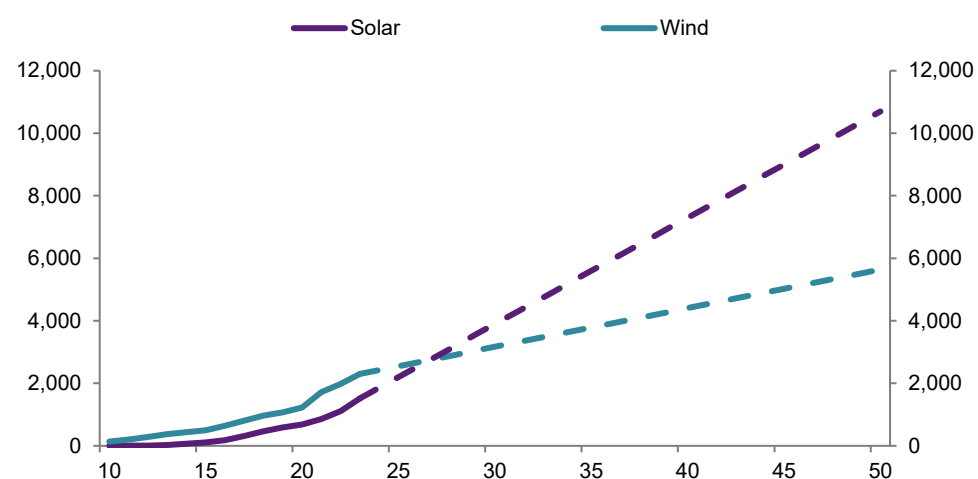
Figure 7: Global primary energy mix (%)



Source: Natixis, Energy Institute, Enerdata.

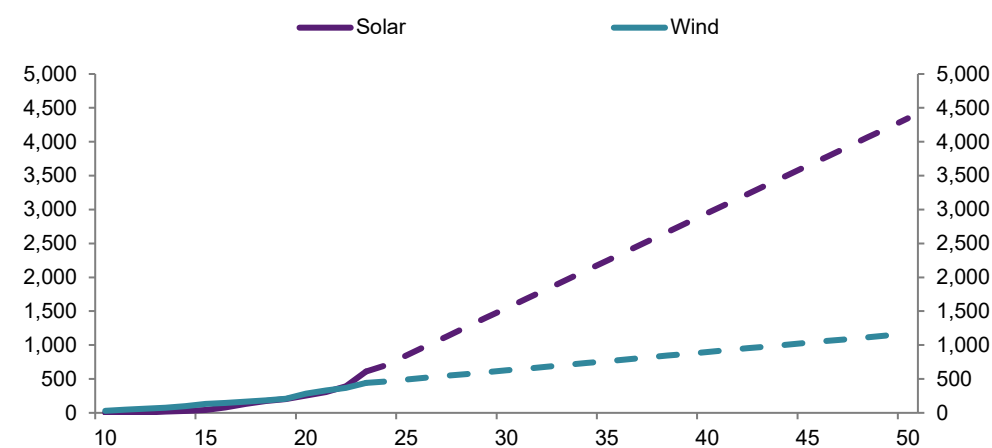
Based on demand for renewable energy, we then calculate the corresponding installations needed by China and the world to fulfil such a transition. For the sake of comparability between renewable energy and traditional fossil fuels, we adopt input-output equivalent conversion from energy demand to renewable power supply, and we assume steadily rising energy efficiency of fossil fuels over the coming decades. We also assume stable utilization rates of solar and wind farms in comparison with the average level during 2021–23. We use linear interpolation for the accumulated installations between now and 2050. We selected the highest forecasts for China's solar and wind energy demand, namely the IEA's Announced Pledges Scenario (APS) and BP's Net Zero Scenario (Raimi et al. 2024). For the rest of the world, we use the IEA's Net Zero Emission (NZE) by 2050 scenario, as it forecasts the highest renewable energy demand among all scenarios. The projection results are shown below.

Figure 8: China renewable energy demand (TWh)



Source: Natixis, IEA APS, BP Net Zero, Authors' calculation.

Figure 9: China renewable power capacity (GW)

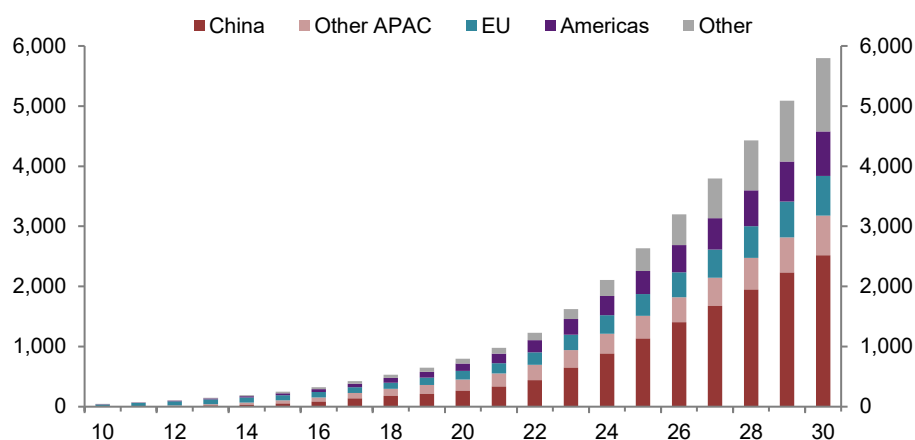


Source: Natixis, NBS, Authors' calculation.

From the trends above, it is not difficult to see that, in order to fulfil its energy transition roadmap, China could slow down its pace and still deliver the transition, whereas the rest of the world will need to catch up in installation speed, as it is behind schedule, which matches our findings from the energy demand projections. This means China's currently fast-growing demand for domestic renewable installations is likely to normalize in the coming years. Meanwhile, other countries will need even more renewable products to fulfil the transition on time.

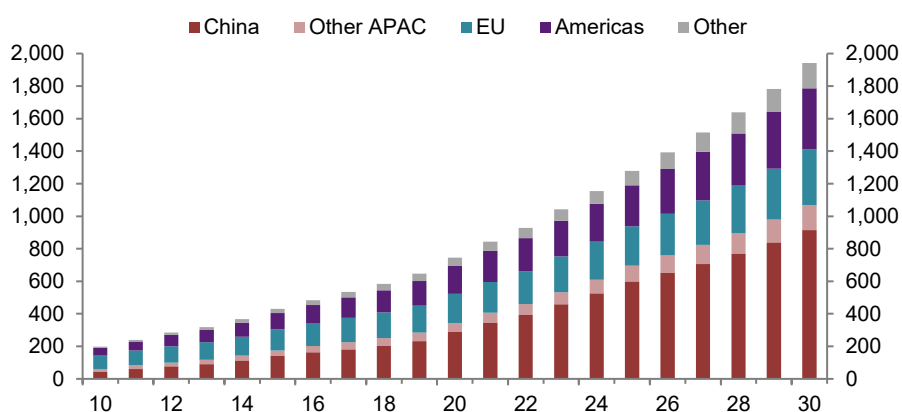
4.3 China's production capacity

Figure 10: Forecast of global solar capacity (GW)



Source: Natixis, BNEF.

Figure 11: Forecast of global wind capacity (GW)



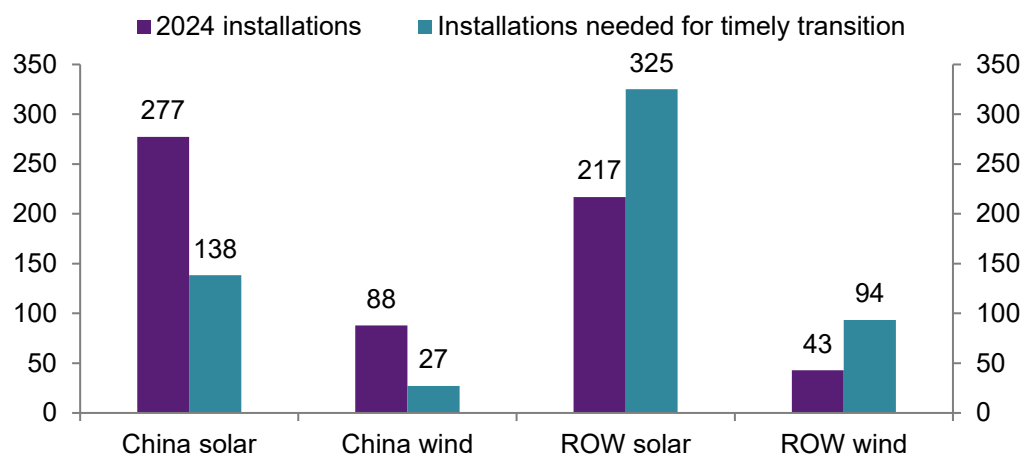
Source: Natixis, BNEF.

China's existing domestic renewable capacity is already the largest in the world by a significant margin. With 36% of the world's total installed capacity, China possesses more solar capacity than the US, the EU, and the rest of the Asia-Pacific region combined. This single fact underscores a critical reality: the world is already heavily reliant on China's green technology ecosystem. Dependency is not merely on future production but is rooted in the existing industrial and energy landscape, a direct result of China's long-term, state-driven focus on the sector.

A direct comparison of current installation rates versus the rates required for a timely transition reveals a critical divergence in investment and deployment priorities. China is currently installing solar capacity at 277 GW annually, more than 2.5 times the 138 GW needed to sustain the Paris Agreement path. It is effectively overinvesting in domestic capacity relative to its needs. The ROW, conversely, installs 217 GW against a required 325 GW, indicating that it is already falling behind

schedule. This gap is even more pronounced in wind power. This dynamic of overbuilding in China and underbuilding elsewhere is a primary driver of China's expanding manufacturing role for the world.

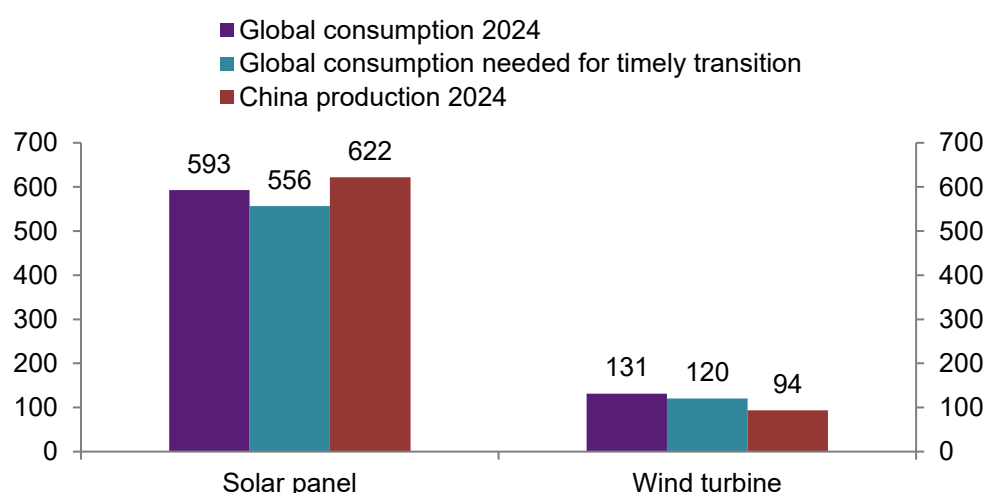
Figure 12: Current annual installed capacity vs. decarbonization demand (GW)



Source: Natixis. Timely transition refers to limiting temperature rise compared to pre-industrialization below 2C by 2100.

The ultimate consequence of China's industrial strategy is a manufacturing capacity that structurally outstrips global needs. In the solar sector, China's annual production capacity of 622 GW is more than 150 per cent of the entire world's required annual installation of 404 GW. For wind turbines, China's production of 94 GW is nearly sufficient to cover the entire global need of 108 GW. This supply–demand imbalance, hardwired into the global market by China's industrial scale, makes its role as the world's primary green tech supplier an economic inevitability.

Figure 13: Global annual demand vs. Chinese production (GW)



Source: Natixis. Timely transition refers to limiting temperature rise compared to pre-industrialization below 2C by 2100.

The data presents an unambiguous picture. A profound structural imbalance defines the global green technology landscape, born from two distinct trends: China's policy driven domestic over investment and the rest of the world's laggardly pace of deployment. China's manufacturing apparatus, scaled to meet its own ambitious targets, is now an overwhelming force on the world stage. This makes global reliance on Chinese production for the energy transition less a matter of choice and more a matter of economic reality, setting the stage for significant geopolitical and industrial consequences.

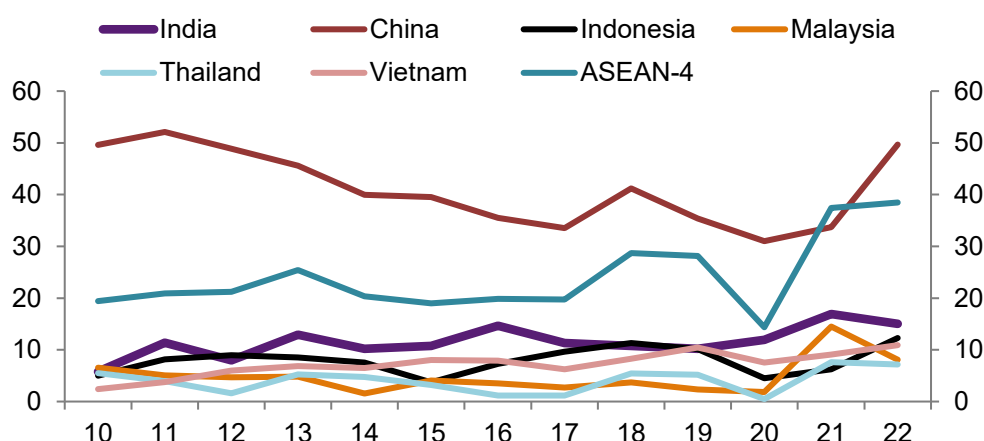
5 How did China manage to produce so much?

China's manufacturing supremacy in green technology is not accidental. It is the result of a deliberate multi-pronged industrial strategy executed over more than a decade. Strategy rests on three core pillars: strategic acquisition of foreign technology, massive and sustained state subsidies, and a hyper-competitive domestic market. This section deconstructs this model to explain how China built its formidable production capacity.

5.1 Strategic technology acquisition

China's initial strategy centred on leveraging foreign direct investment (FDI) for technology transfer (Zhou and Jiang 2020). State policy created a vast protected domestic market, making participation irresistible for foreign firms. Access to this market was often implicitly conditional on forming joint ventures and sharing technology with local partners. This market-for-technology strategy, seen clearly in the automotive and wind turbine sectors, allowed Chinese firms to rapidly absorb and localize foundational technologies. This effectively enabled them to leapfrog years of incremental R&D and quickly reach the technological frontier.

Figure 14: Manufacturing FDI inflow (USDbn)



Source: Natixis, CEIC.

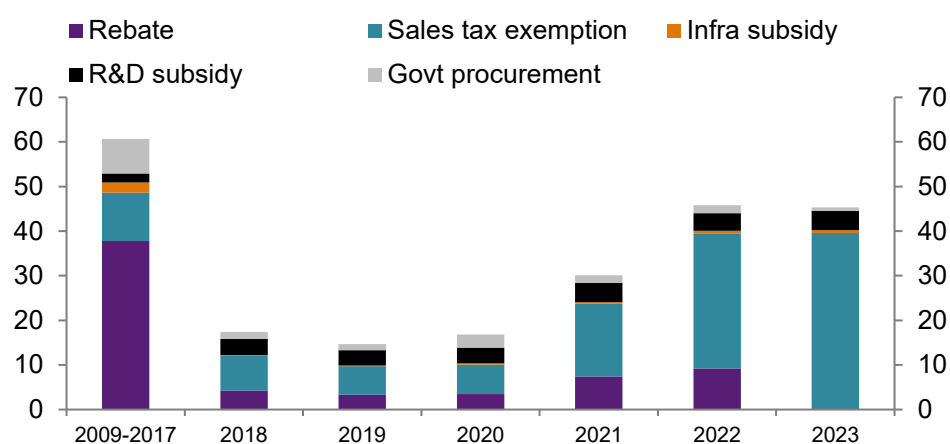
5.2 Massive state subsidization

The most critical element has been extensive and sustained government support. The electric vehicle (EV) sector is a prime example. Through policies such as the 'Ten Cities, One Thousand Vehicles'

programme launched in 2009, China has spent massively to subsidize the entire value chain. It financed supply-side production while simultaneously stimulating demand with consumer rebates (Zheng et al. 2022). This dual-pronged fiscal support created a market and an industry simultaneously, underwriting the sector's explosive growth.

Support is persistent and adaptable. The state maintained high levels of support even as direct purchase rebates were phased out in 2023. Significant sales tax exemptions for consumers remain a powerful demand side stimulus. Moreover, in response to economic headwinds, the government has reintroduced subsidies through large-scale equipment renewal and vehicle trade-in schemes, demonstrating a pragmatic and unwavering commitment to the sector.

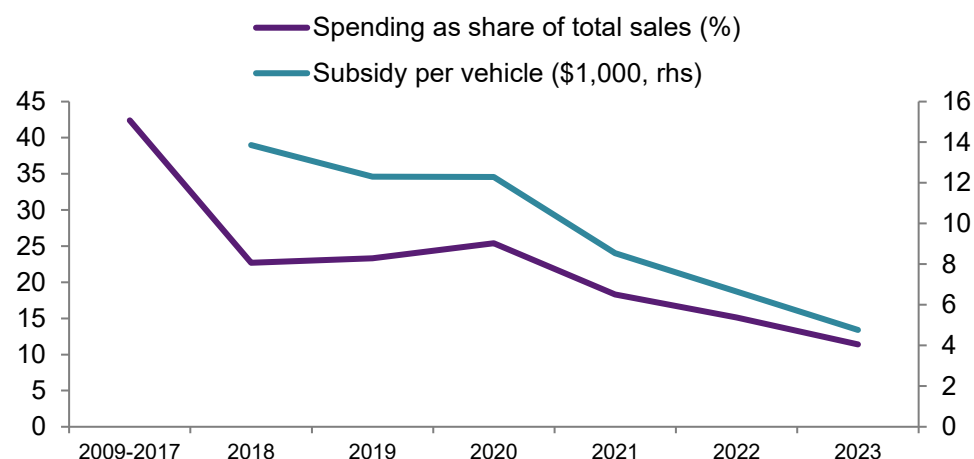
Figure 15: Industry policy spending for China's EV sector (USDbn)



Source: Center for Strategic and International Studies, Natixis.

Although price rebate has been canceled in 2023, sales tax exemption remains high. Moreover, due to China's economic headwinds, China bring back the large-scale renewal of equipment project. Under the trade-in scheme, consumers can receive a subsidy of up to RMB 20,000 when they scrap an older car.

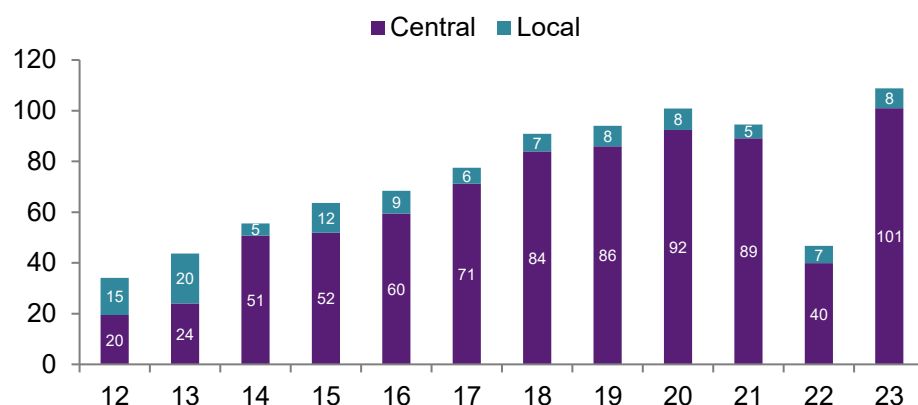
Figure 16: Industry policy spending to sales and per vehicle



Source: Center for Strategic and International Studies, Natixis.

The subsidy model extends across the entire green tech landscape. China began subsidizing electricity from renewable sources in 2006 through a national feed-in tariff system. This policy was crucial for closing the price gap with established thermal power. It made wind and solar projects economically viable and bankable, attracting enormous investment. While subsidies for some large-scale projects have declined, spending rebounded in 2023, driven by a new focus on distributed household solar devices.

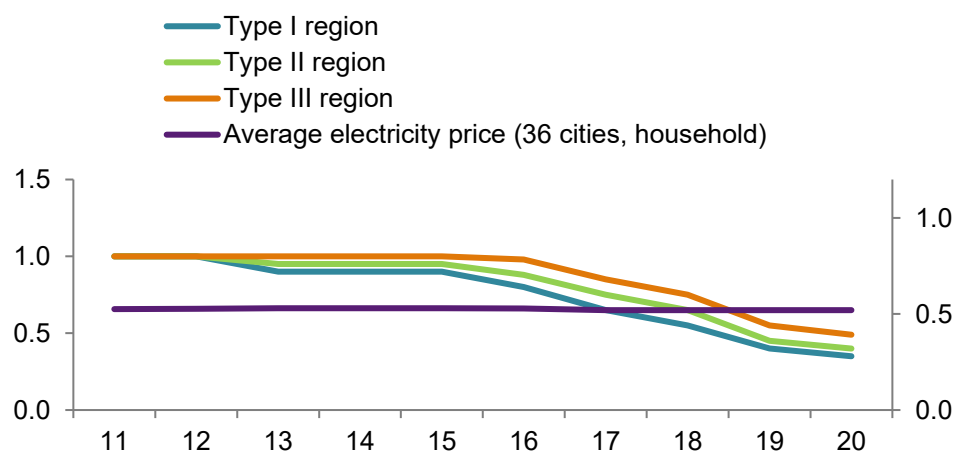
Figure 17: Subsidy on renewable electricity price (RMBbn)



Source: Natixis, Ministry of Finance.

The correlation between state support and cost reduction is direct and unambiguous. The average solar power price plummeted as subsidies ramped up. State capital was used to drive the industry to a scale at which it became cost-competitive on its own merits. This is a key objective of strategic industrial policy.

Figure 18: Solar power price (yuan/kWh, grid*)

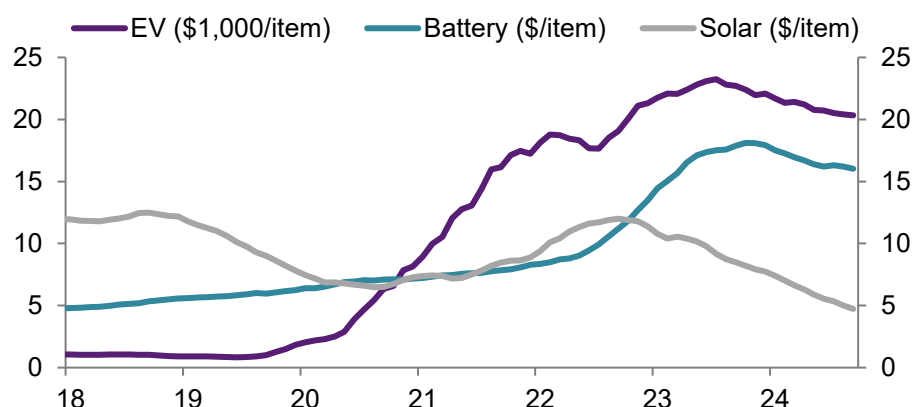


Source: Natixis, NDRC. *From 2021 on, solar electricity is not subject to subsidy for centralized projects and distributed projects for industrial and commercial purpose.

5.3 Hyper competition

The third pillar of China's strategy is the paradox of state-managed hyper-competition. Alongside massive state support, the government has allowed and even encouraged ferocious domestic competition. A flood of capital into the sector, from both state and private sources, created a winner-takes-all dynamic. This fostered intense price wars among a vast number of domestic firms, a process the state viewed as necessary to weed out inefficiency and forge globally competitive national champions. The result is a sharp and continuous decrease in the export prices of EVs, batteries, and solar panels.

Figure 19: China average export price (12mma)

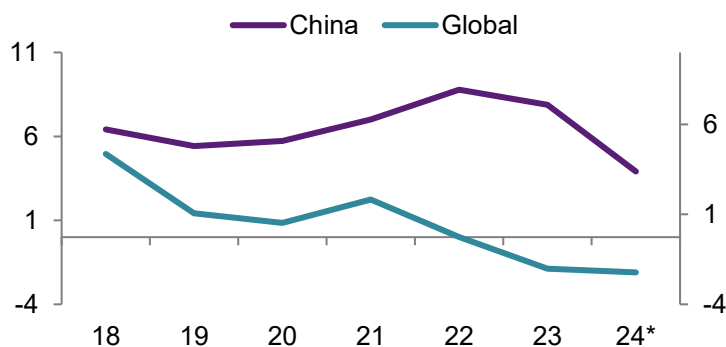


Note: Data as of September 2024.

Source: Natixis, China General Administration of Customs.

Hyper-competition has had a direct and severe impact on corporate balance sheets. The relentless pressure to cut prices in order to gain market share has led to a significant compression of profit margins for Chinese green tech companies. This is a classic sign of an overcrowded and highly competitive market. The inevitable consequence of falling profitability is a future of market consolidation and reduced capital expenditure (capex). This signals the end of the hyper-growth phase. The industry is now entering a new stage of maturation that will be defined by widespread consolidation and the emergence of a few dominant players.

Figure 20: Listed renewable companies: profit margin (%)



Note: Data as of 2024 Q2 (Trailing 12-month). Two-year average.

Source: Natixis, Financial Statements, *Bloomberg*.

6 Consequences for the global economy

The success of China's green technology sector, driven by its unique industrial development model, has created significant and complex consequences for the global economy. The sheer scale of its production now spills over into international markets, functioning as a powerful economic force and presenting a dual-edged reality for the rest of the world. On the one hand, China's capacity offers a historic opportunity to accelerate the global energy transition. On the other hand, it poses a formidable competitive challenge that is forcing a fundamental reassessment of national industrial strategies worldwide.

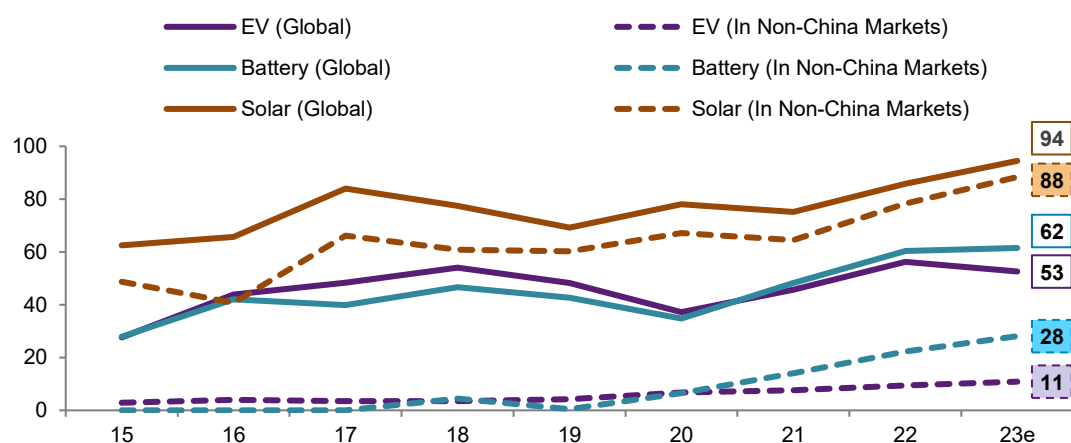
6.1 The competitive challenge and industrial policy

The competitive advantage held by Chinese firms is not rooted in any single policy but in the comprehensive and integrated nature of the state's entire industrial strategy. Other nations find it difficult to compete not simply because of the scale of subsidies, but because they cannot easily replicate the entire ecosystem. This model combines a vast, protected domestic market that serves as a test bed; long-term, unwavering policy signals such as the 'Dual Carbon' goals that de-risk private investment; state-directed capital allocation; and a managed tolerance for intense domestic competition that drives down costs.

The result is China's rapid capture of overseas markets. In the solar panel sector, China supplied 88% of the global market in 2023. Its rise in newer industries is just as swift. Its market share in non-China markets for EVs has surged to 28% and for batteries to 11%, up from almost nothing just five years prior.

This integrated approach creates a structural competitive advantage that is deeply embedded in China's political economy. For other countries, with different market structures and political systems, deploying a similarly holistic and state-directed industrial policy on such a scale is often not feasible. This reality is forcing a strategic recalculation in Western capitals. The response is a wave of new industrial policies, such as the Inflation Reduction Act in the United States and the EU's Green Deal Industrial Plan. These initiatives represent attempts to build resilient domestic supply chains, but they face the immense challenge of competing with an established and highly efficient Chinese industrial base. This creates a fundamental policy dilemma for governments: either accept a degree of dependency on China's low-cost supply or commit to costly and complex long-term strategies to foster domestic alternatives.

Figure 21: China market share in green tech (%)

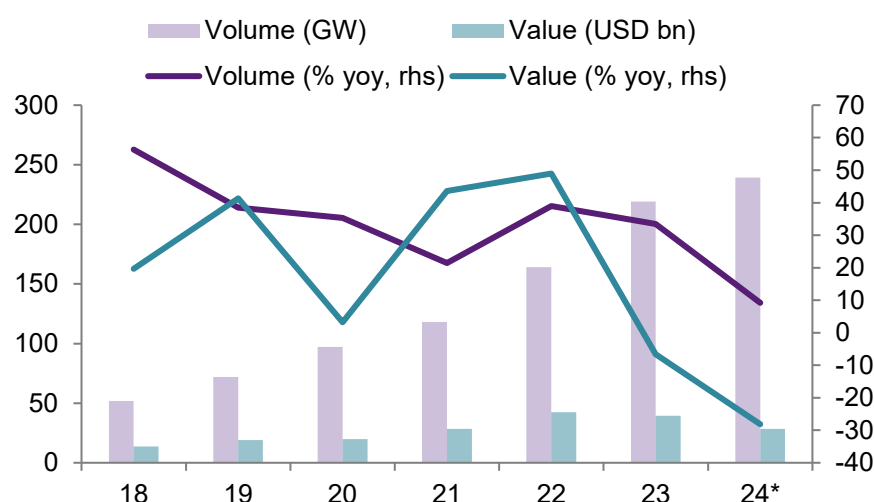


Source: Natixis, MarkLines, BloombergNEF, SNE Research.

6.2 Still opportunity for the rest of world

The hyper-competition within China directly translates into exporting deflationary pressures worldwide. As Chinese firms vie for global market share, they engage in aggressive price-cutting. The solar industry is the most prominent example. As China's export capacity for solar panels has exploded, the price per watt has collapsed.

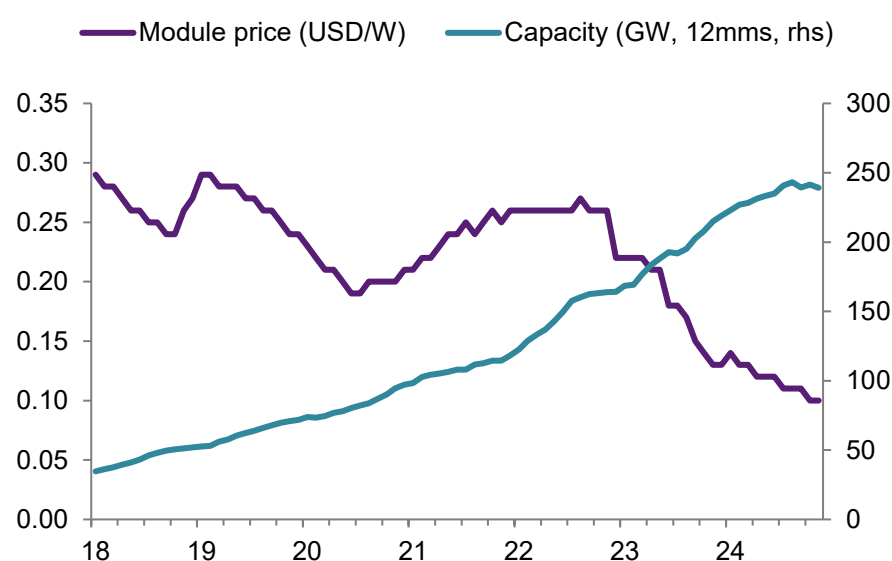
Figure 22: China solar panel export



Source: Natixis, Ember. *12-month moving sum as of November 2024.

From a global welfare perspective, this is a significant positive externality. China's industrial engine is effectively subsidizing the world's energy transition. The availability of low-cost equipment makes decarbonization more economically viable and politically palatable for all nations, particularly for developing countries where cost is a primary barrier to adoption. Accelerated deployment of clean energy is a global public good, helping the international community move closer to its collective climate goals. While this dynamic creates undeniable challenges for competing manufacturers in other countries, it simultaneously provides a powerful tailwind for global climate action. For China itself, the process is not without cost. The domestic price wars put immense pressure on corporate profitability and financial stability, a necessary internal adjustment that is likely to culminate in a significant consolidation of its own green industries.

Figure 23: China solar panel export



Source: Natixis, Ember.

7 Strategic pathways for the Global South

It is increasingly evident that competing with China as a standalone entity is economically unfeasible, given China's dominant position in green technology and its immense government support. The comprehensive nature of China's industrial policy has created a juggernaut that no individual developing nation can easily replicate or challenge. For the Global South, regional alliances offer the only viable mechanism to navigate this new industrial landscape. By aggregating resources and market demand, developing nations can build resilient value chains that complement, rather than directly compete with, China. ASEAN can be used as an example of how such a plan can be conducted.

7.1 Regional alliances and market integration

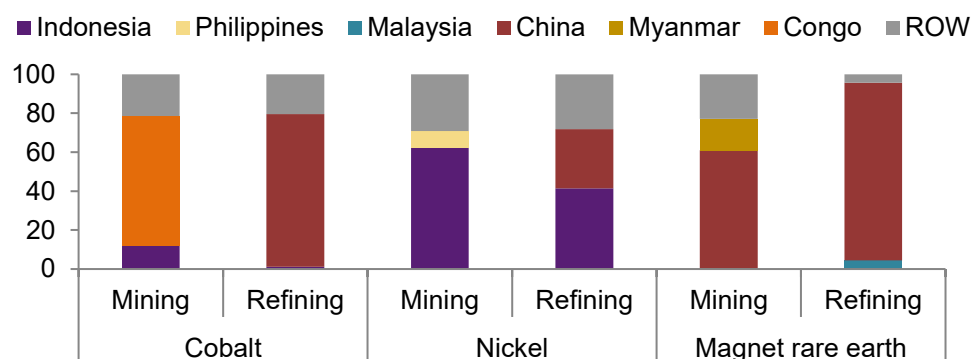
The foundational step for the Global South is the creation of a unified economic bloc capable of attracting substantial FDI. To attract global brands to build supply chains or to foster domestic champions, the prerequisite is a large, accessible domestic market similar to that which China possesses. China's industrial ascent was underpinned by its ability to offer a massive internal market that allowed firms to achieve rapid economies of scale. Because individual Global South economies are relatively small, forming a regional alliance to generate a closed, large-scale market is crucial.

By integrating their markets through agreements that lower tariffs and harmonize regulatory standards, nations can create a collective market size that justifies large-scale capital expenditure. This regional approach transforms fragmented markets into a unified economic bloc. It is this aggregated demand that signals to foreign investors that the region is not merely a collection of small outposts, but a sustainable hub for long-term industrial growth.

7.2 Leveraging critical minerals

Once the market framework is established, the region must leverage its comparative advantage in upstream resources to secure a position in the global value chain. The second step is to secure and refine the critical minerals necessary for the green transition. Indonesia, for instance, holds a dominant share of global nickel reserves, which are essential for EV batteries. By implementing policies that mandate domestic processing and refining, Indonesia has successfully attracted billions in investment for downstream refining facilities. The region can capture significantly more value from its natural resources rather than simply acting as an extractive periphery. Furthermore, securing these critical materials is vital for global supply chain resilience. It provides a strategic export opportunity that advanced economies will rely on, thereby increasing the geopolitical and economic leverage of the regional bloc.

Figure 24: Share of world critical minerals supply (% , 2024)



Note: Data as of 2024.

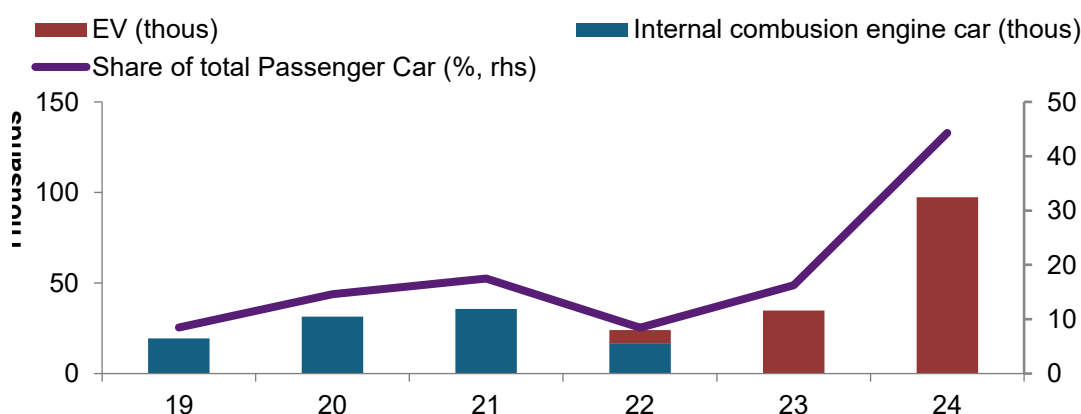
Source: Natixis, IEA.

7.3 Manufacturing integration

The final and most difficult phase is the transition from resource processing to high-value green manufacturing, which requires supporting domestic or regional production through aggressive policy tools that stimulate demand. Governments can deploy tax incentives, registration fee exemptions, and subsidies to boost the adoption of EVs and green technologies, as seen in Thailand and Indonesia. However, the ultimate goal is the creation of full manufacturing capabilities.

Viet Nam stands out as a primary example of this trajectory through the rise of VinFast. Unlike other regional players that focused primarily on attracting foreign automakers, Viet Nam fostered a homegrown national champion. By leveraging a combination of robust corporate strategy and government support, including tax holidays and infrastructure development, VinFast was able to rapidly move from concept to mass production, exporting EVs to global markets including North America. VinFast demonstrates that with the right combination of market access and industrial policy, a Global South nation can move beyond assembly and successfully nurture complex, proprietary manufacturing brands. This builds not only industrial capacity but also creates a specialized export base to serve other regions.

Figure 25: VinFast automobile sales



Source: Natixis, Vingroup Annual Report, CEIC.

8 Conclusions

China's ascendancy in the green technology sector represents a fundamental shift in the global industrial order, driven by an existential domestic imperative to secure energy independence. This state-directed strategy has created an ecosystem of unprecedented scale that positions China as the indispensable supplier for the global energy transition. While this lowers decarbonization costs for the world, it also creates acute risks of industrial dependency for developing nations. The era of competing as fragmented markets is therefore over. The viable pathway for the Global South requires the construction of robust regional alliances that aggregate demand so as to justify large-scale investment. By leveraging this unified market power to enforce domestic processing of critical minerals, developing nations can cultivate their own manufacturing capabilities. This sequential strategy moves the region from passive consumption to active industrial participation, ensuring that they thrive alongside China in the green revolution.

References

- Chen, Y., Chen, G. and Xu, W. (2025). 'Analysis of the impact of outward foreign direct investment of corporations on green innovation in the context of the Belt and Road Initiative'. *Sustainability*, 17(9), pp. 3773–3773. <https://doi.org/10.3390/su17093773>
- Cheng, Z., Yu, X. and Zhang, Y. (2023). 'Is the construction of new energy demonstration cities conducive to improvements in energy efficiency?' *Energy*, 263, p. 125517. <https://doi.org/10.1016/j.energy.2022.125517>
- He, J., Xue, H., Yang, W., Zhong, Y. and Fan, B. (2025). 'Green credit policy and corporate green innovation'. *International Review of Economics & Finance*, 99, p. 104031. <https://doi.org/10.1016/j.iref.2025.104031>
- Kariyawasam, K. and Tsai, M. (2018). 'Intellectual property, climate change and technology transfer in South Asia'. In: Rimmer, M. (ed.), *Intellectual Property and Clean Energy*. Singapore: Springer, pp. 207–234. https://doi.org/10.1007/978-981-13-2155-9_8
- Korabayev, R. and Korabayev, K. (2023). 'Development of renewable energy businesses within China's green industry framework'. *Management and Production Engineering Review*, 14(4). <https://doi.org/10.24425/mper.2023.147209>
- Ma, Y. (2025). 'The impact of the new energy demonstration city policy on green growth in pilot cities: evidence from a quasi-experiment in Chinese prefecture-level cities'. *Journal of Innovation and Development*, 11(3), pp. 84–98. <https://doi.org/10.54097/60k2eq05>
- Maji, S.K. and Laha, A. (2021). 'The role of digital skill in mitigating digital divide: evidences from Asia-Pacific region'. *Rajagiri Management Journal*, 16(3), pp. 260–271. <https://doi.org/10.1108/RAMJ-05-2021-0035>
- Nepal, R., Phoumin, H. and Khatri, A. (2021). 'Green technological development and deployment in the Association of Southeast Asian Economies (ASEAN): at crossroads or roundabout?' *Sustainability*, 13(2), p. 758. <https://doi.org/10.3390/su13020758>
- Peimani, H. (2018). 'Financial barriers to development of renewable and green energy projects in Asia'. Working Papers. Available at: <https://ideas.repec.org/p/ess/wpaper/id12891.html>
- Qudrat-Ullah, H. (2023). 'A review and analysis of green energy and the environmental policies in South Asia'. *Energies*, 16(22), pp. 7486–7486. <https://doi.org/10.3390/en16227486>
- Raimi, D., Zhu, Y., Newell, R.G. and Prest, B.C. (2024). *Global Energy Outlook 2024: Peaks or Plateaus?* Washington, DC: Resources for the Future. Available at: https://media.rff.org/documents/Report_24-06.pdf
- Shi, S. (2023). 'Research on the impact mechanism of government subsidies on green innovation of enterprises'. *Advances in Economics and Management Research*, 6(1), p. 653. <https://doi.org/10.56028/aemr.6.1.653.2023>
- Sun, G., Feng, Q., Wu, S. and Xia, C. (2025). 'The impact of China's green credit policy on the innovation of manufacturing enterprises'. *Journal of Innovation & Knowledge*, 10(3), p. 100714. <https://doi.org/10.1016/j.jik.2025.100714>

- Veugelers, R. (2014). 'What innovation policies for ecological transition? Powering the green innovation machine'. Available at: <http://hdl.handle.net/10419/125726>
- Wang, S. and Hao, J. (2012). 'Air quality management in China: issues, challenges, and options'. *Journal of Environmental Sciences*, 24(1), pp. 2–13. [https://doi.org/10.1016/S1001-0742\(11\)60724-9](https://doi.org/10.1016/S1001-0742(11)60724-9)
- Yang, G. (2023). 'The technological innovation capability of the new energy vehicle industry'. *E3S Web of Conferences*, 385, pp. 01029–01029. <https://doi.org/10.1051/e3sconf/202338501029>
- Zheng, X., Menezes, F., Zheng, X. and Wu, C. (2022). 'An empirical assessment of the impact of subsidies on EV adoption in China: A difference-in-differences approach'. *Transportation Research Part A: Policy and Practice*, 162, pp. 121–136. <https://doi.org/10.1016/j.tra.2022.05.020>
- Zhou, W. and Jiang, H. (2020). 'Technology transfer under China's foreign investment regime: Does the WTO provide a solution?' *Journal of World Trade*, 54(3). <https://doi.org/10.54648/TRAD2020021>